

Wi-Fi Series

Data Sheet

USE_8800

Wi-Fi / BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	04/08/2026	MB	LB	Initial release

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Features :

General

- Support Wi-Fi/Bluetooth functionalities
- Compliant with IEEE 802.11a/b/g/n/ac/ax
- Bluetooth 5.4 compliant
- Support Wi-Fi 6 TWT
- 1T1R architecture
- Support STBC, beamforming
- Dimension: 12 x 12 x 1.7 mm
- Operating temperature: -20°C to 80°C
- Storage temperature: -40°C to 125°C
- OS supported : Android /Linux

MCU Platform

- Chipset: AIC8800D40

WLAN Features

- Support 2.4G+5G dual band, 20/40MHz bandwidth
- Support STA, SoftAP, Wi-Fi Direct modes concurrently
- Data rates up to 286.8Mbps with 20/40MHz bandwidth
- Wi-Fi Security WEP / WPA / WPA2/WPA3-SAE Personal, MFP

Host Interface

- Supports SDIO3.0 interface for WLAN
- Supports UART interface for Bluetooth
- Supports SDIO3.0/USB2.0

Bluetooth Features

- Supports all the mandatory and optional features of Bluetooth 5.4
- Supports advanced master and slave topologies
- Small SMT package

General Description:

The **USE_8800** is a highly integrated chip with dual-band Wi-Fi 6 and Bluetooth 5.4 for wireless applications. Its WLAN function supports the SDIO 3.0 interface, and Bluetooth supports the UART interface only. The module provides a simple legacy and 20MHz/40MHz co-existence mechanism to ensure backward and network compatibility, and supports both 2.4GHz and 5GHz Wi-Fi 6 bands.

The wireless module complies with the IEEE 802.11a/b/g/n/ac/ax standard and supports data rates up to 286.8Mbps with 20/40MHz bandwidth operation.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

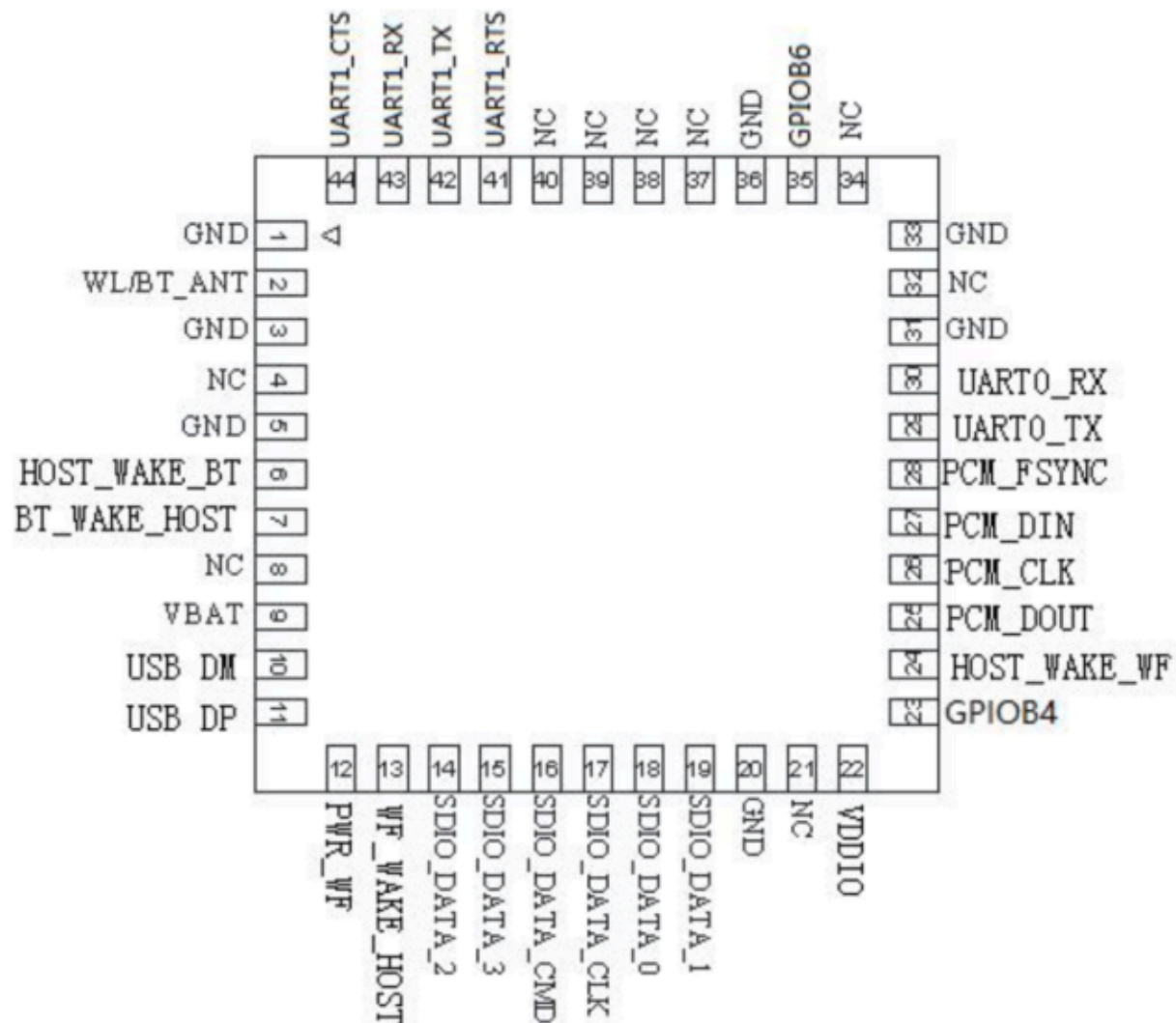


Fig1: Module Pinouts

Pin Descriptions:

NO.	Name	Type	Description	Voltage
1	GND	—	Ground connections	
2	WL/BT_ANT	I/O	WLAN and BT RF I/O port	
3	GND	—	Ground connections	
4	NC	—	Floating (Don't connected to ground)	
5	GND	—	Ground connections	
6	HOST_WAKE_BT	I/O	GPIOB3/HOST_WAKE_BT	VIO
7	BT_WAKE_HOST	I/O	GPIOB2/BT_WAKE_HOST	VIO
8	NC	—	Floating (Don't connected to ground)	
9	VBAT	P	Main power voltage source input	3.3V
13	WF Wake Up Host	I/O	WF Wake Up Host	VIO
14	SD_D2	I/O	SDIO data line 2, GPIOA15	VIO
15	SD_D3	I/O	SDIO data line 3, GPIOA14	VIO
16	SD_CMD	I/O	SDIO command line, GPIOA13	VIO
17	SD_CLK	I/O	SDIO clock line ,GPIOA12	VIO
18	SD_D0	I/O	SDIO data line 0 ,GPIOA11	VIO
19	SD_D1	I/O	SDIO data line 1, GPIOA10	VIO
20	GND	—	Ground connections	
21	NC	—	Floating (Don't connected to ground)	
22	VDDIO	P	I/O Voltage	1.8V/3.3V
23	GPIOB4	I/O	GPIOB4,if not used please keep NC	VIO
24	HOST_WAKE_WF	I/O	HOST_WAKE_WF	VIO
25	PCM_DOUT	O	PCM Data output	VIO
26	PCM_CLK	I/O	PCM clock	VIO
27	PCM_DIN	I	PCM data input	VIO
28	PCM_SYNC	I/O	PCM sync signal	VIO
29	UART0 TX	I/O	GPIOA9, UART0 TX	VIO
30	UART0 RX	I/O	GPIOA8, UART0 RX	VIO
31	GND	—	Ground connections	
32	NC	—	Floating (Don't connected to ground)	
33	GND	—	Ground connections	
34	NC	—	Floating (Don't connected to ground)	

35	GPIOB6	I/O	GPIOB6,,if not used please keep NC	VIO
36	GND	—	Ground connections	
37	NC	—	Floating (Don't connected to ground)	
38	NC	—	Floating (Don't connected to ground)	
39	NC	—	Floating (Don't connected to ground)	
40	NC	—	Floating (Don't connected to ground)	
41	UART1_RTS	I/O	BT_UART1_RTS, GPIOA7	VIO
42	UART1_TX	I/O	BT_UART1_TX, GPIOA5	VIO
43	UART1_RX	I/O	BT_UART1_RX, GPIOA4	VIO
44	UART1_CTS	I/O	BT_UART1_CTS, GPIOA6	VIO

The signal type codes are used in the table:

- P:POWER
- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	-20	25	80	°C
VBAT	3.0	3.3	3.6	V
VDDIO	1.7	1.8 / 3.3	3.6	V

Power Consumption

Test condition: VBAT = 3.3 V / VDDIO = 3.3 V

2.4 GHz

Operating Mode	Maximum TX Current (mA)	Maximum RX Current (mA)
11b @ 20 dBm	268	57
11g @ 17 dBm	187	57
HT20-MCS7 @ 16 dBm	179	58
HT40-MCS7 @ 16 dBm	163	57

VHT20-MCS8 @ 16 dBm	164	56
VHT40-MCS9 @ 16 dBm	162	56
HE20-MCS11 @ 15 dBm	138	57
HE40-MCS11 @ 15 dBm	134	58

5GHz

Operating Mode	Maximum TX Current (mA)	Maximum RX Current (mA)
54M @ 17 dBm	215	58
MCS7 HT20 @ 17 dBm	206	56
MCS7 HT40 @ 17 dBm	188	56
MCS8 VHT20 @ 17 dBm	203	58
MCS9 VHT40 @ 16 dBm	186	57
MCS11 HE20 @ 15 dBm	185	58
MCS11 HE40 @ 14 dBm	174	58

RF Characteristics

1. 2.4G RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11b/g/n/ax, Wi-Fi compliant	
Frequency Range	2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band)	
Number of Channels	2.4GHz : Ch1 ~ Ch13,Ch14	
Modulation	DBPSK/DQPSK/CCK(DSSS)、BPSK/QPSK/16QAM/64QAM(OF DM)	
Test Items	Typical Value	EVM
Output Power	802.11b /11Mbps : 18dBm $\pm$ 2 dBm	EVM -10dB
	802.11g /54Mbps : 15dBm $\pm$ 2 dBm	EVM -25dB
	802.11n /MCS7: 15dBm $\pm$ 2 dBm	EVM -28dB
	802.11ax /MCS9 :14dBm $\pm$ 2 dBm	EVM -32dB
	802.11ax /MCS11 :13dBm $\pm$ 2 dBm	EVM -35dB

Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	±20ppm	
ReceiveSensitivity (11b,20MHz) @8% PER	- 11Mbps@-85dBm	≤-76dBm
ReceiveSensitivity (11g,20MHz) @10% PER	- 54Mbps@-73dBm	≤-68dBm
Receive Sensitivity (11n,20MHz) @10% PER	- MCS=7 @-70dBm	≤-67dBm
Receive Sensitivity (11n,40MHz) @10% PER	- MCS=7 @-68dBm	≤-64dBm
Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=11@-63dBm	≤-52dBm
Receive Sensitivity(11ax,40MHz) @10% PER	- MCS=11@61-dBm	≤-49dBm
Maximum Input Level	802.11b : -10 dBm	
	802.11g/n : -20 dBm	
Antenna Reference	Small antennas with 0~2 dBi peak gain	

2. 5GHz RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11 a/n Wi-Fi compliant	
Frequency Range	5.150 GHz ~ 5.850 GHz (5.0 GHz Band)	
Test Items	Typical Value	EVM
Output Power	802.11a 54Mbps: 15dBm ± 2 dBm	EVM -25dB
	802.11n MCS7: 15 dBm± 2 dBm	EVM -28dB
	802.11ac MCS8: 14dBm ± 2 dBm	EVM -30dB
	802.11ac MCS9: 14dBm ± 2 dBm	EVM -32dB
	802.11ax MCS11 : 12dBm ± 2 dBm	EVM -35dB
ReceiveSensitivity (11a,20MHz) @10% PER	- 6Mbps PER @ -89 dBm, typical	≤-85
	- 54MbpsPER @ -71 dBm, typical	≤-68
Receive Sensitivity	- MCS=0 PER @ -89 dBm, typical	≤-85

(11n,20MHz) @10% PER	- MCS=7 PER @ -69 dBm, typical	≤-67
Receive Sensitivity	- MCS=0 PER @ -87 dBm, typical	≤-82
(11n,40MHz) @10% PER	- MCS=7 PER @ -67 dBm, typical	≤-64
Receive Sensitivity	- MCS=0 PER @ -85 dBm, typical	≤-81
(11ac,20MHz) @10% PER	- MCS=8 PER @ -66 dBm, typical	≤-59
Receive Sensitivity	- MCS=0 PER @ -83 dBm, typical	≤-79
(11ac,40MHz) @10% PER	- MCS=9 PER @ -63 dBm, typical	≤-54
Receive Sensitivity	- MCS=0 PER @ -82 dBm, typical	≤-78
(11ax,20MHz) @10% PER	- MCS=11 PER @ -65 dBm, typical	≤-52
Receive Sensitivity	- MCS=0 PER @ -81 dBm, typical	≤-78
(11ax,40MHz) @10% PER	- MCS=11 PER @ -62 dBm, typical	≤-49
Maximum input level	802.11a/n: -30 dBm	
	802.11ac: -30 dBm	
	802.11ax: -30 dBm	

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

3. Bluetooth Specification

Feature	Description
General Specification	
Bluetooth Standard	BDR/EDR(1Mbps/2Mbps/3Mbps),LE(1Mbps) LE(2Mbps)
Host Interface	UART
Frequency Band	2400 MHz ~ 2483.5 MHz
Number of Channels	79 channels for classic 、 40 channels for BLE
Modulation	GFSK, $\pi/4$ -DQPSK, 8-DPSK
RF Specification	
Output Power , tolerance ± 3 dB	
	CL1(dBm)

BDR Output Power	9
EDR Output Power	8
BLE Output Power	9
Sensitivity, tolerance :/ dB	
Sensitivity @ BER=0.1% for GFSK (1Mbps)	-83
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)	-83
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)	-75
Sensitivity @ BER=30.8% for LE (1Mbps)	-85
Sensitivity @ BER=30.8% for LE (2Mbps)	-83
Maximum Input Level	GFSK :-20dBm

Block Diagram :

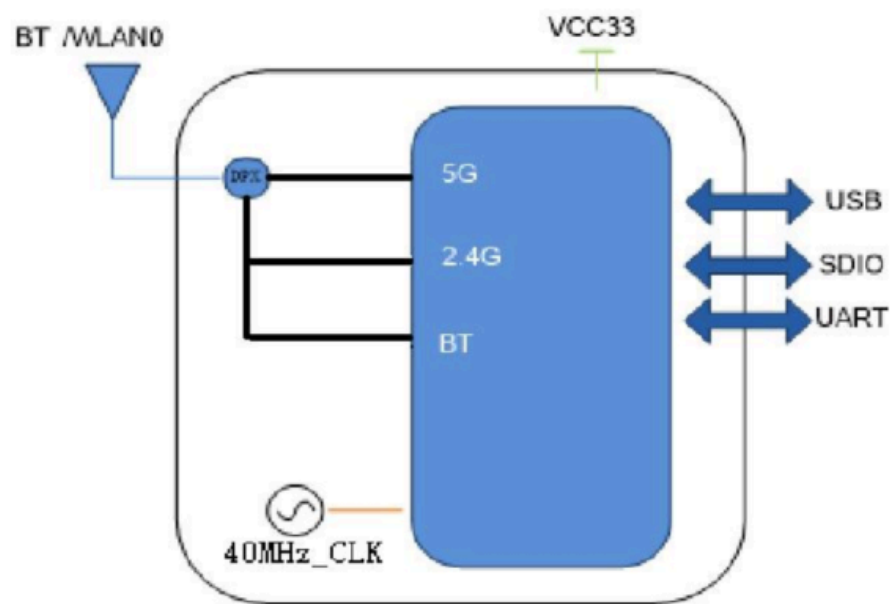


Fig 2 : Block Diagram

Reference Design :

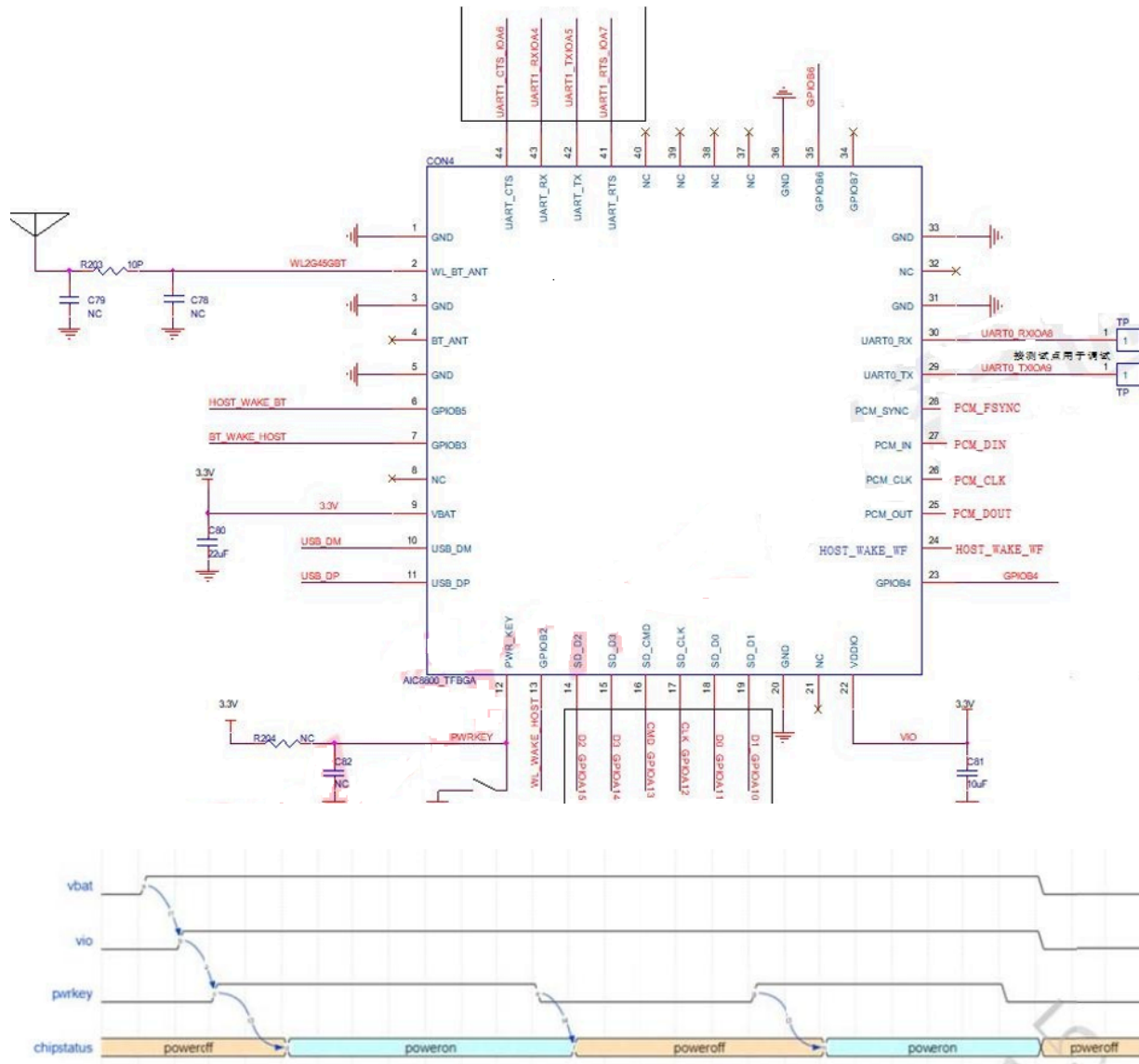


Fig 3 : Reference Design

- t1: VIO's power on time $\geq$ VBAT's
- t2: Power key's high time $\geq$ VIO's +200us
- t3: Chip all power on ready time $\geq$ power key high time + 8ms
- t4: Power key pull low to chip all power off time $\geq$ 6ms

Recommended Footprint Dimensions:

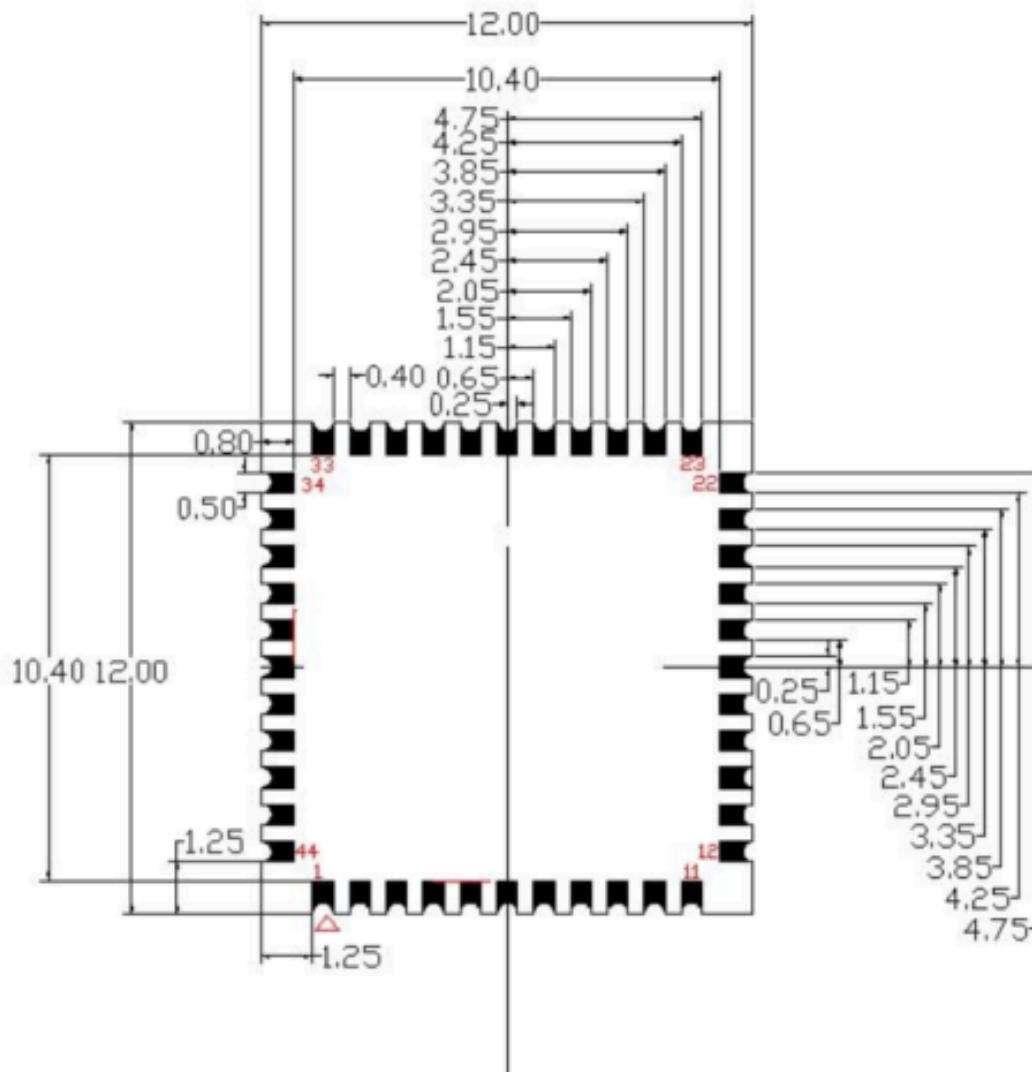


Fig 4(a) : Recommended Footprint Dimensions



Module Dimension Drawing:

- Module Dimension: L x W :12 x 12 (± 0.2) mm

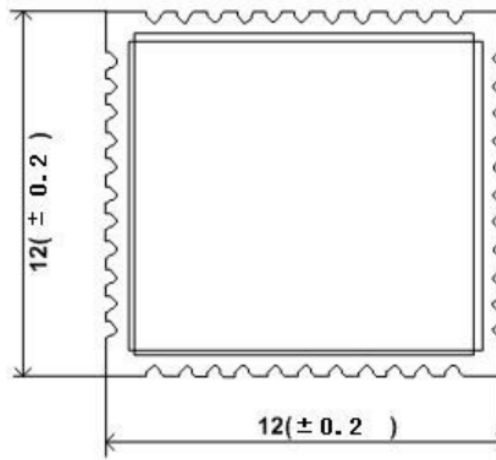


Fig 5(a) Module Dimension L × W

Product Dimension:

- H: 1.7 (± 0.2) mm

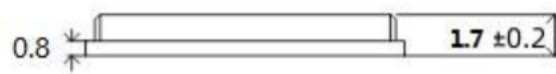


Fig 5(b) :Product Dimension

PCB thickness : 0.8 mm

Product Height : 1.7 ± 0.2 mm

Weight : 0.41(± 0.1)g

Recommended Reflow Profile :

Referred to IPC/JEDEC standard

Peak Temperature : $\leq 260^{\circ}\text{C}$

Time within 5°C of peak

temperature : $\geq 10\text{s}$

Number of Times : ≤ 2 times

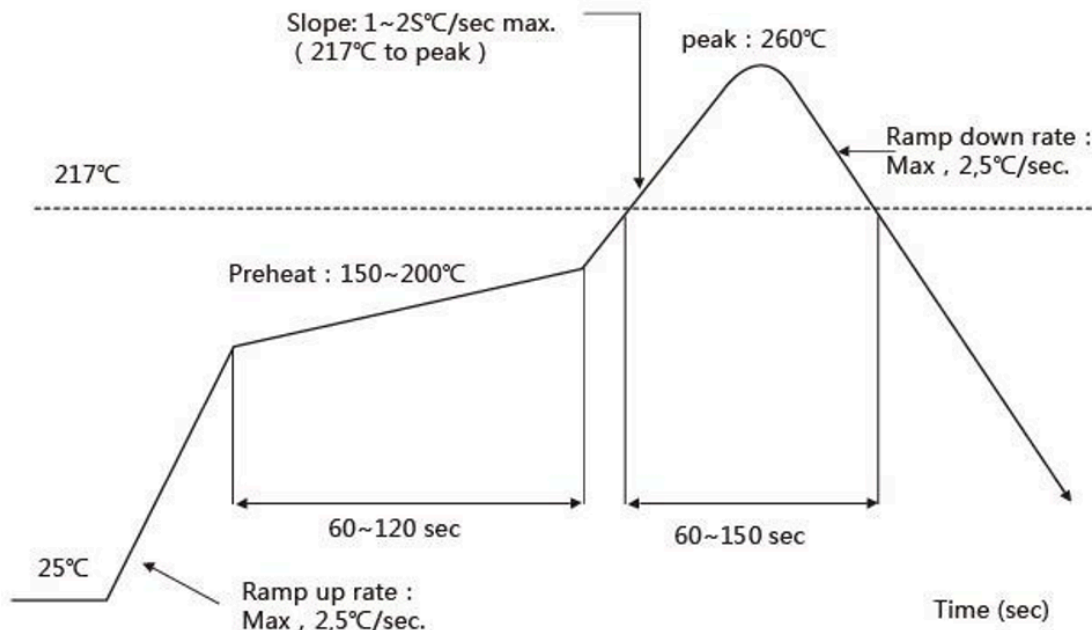
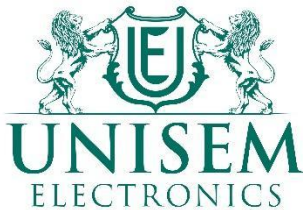


Fig 6 : Recommended Reflow Profile

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